

Intermediate Macro

Chapter 5: The $IS - LM$ Model

Instructor: Mihyun You

Department of Economics
Georgetown University

Summer 2022



GEORGETOWN UNIVERSITY

5.5 How Does the $IS - LM$ Model Fit the Facts?



Key Tools and Concepts

Key Tools and Concepts

- Chapter 5 introduces the *IS-LM* framework.
- Chapter 5 introduces the concept of **policy mixes** to achieve macroeconomic goals.

Assumptions

- Chapter 5 maintains the fixed price assumption of previous chapters.
- We relaxes the assumptions that investment is independent of the interest rate (assumed in Chapter 3) and that nominal income is fixed (assumed in Chapter 4).
- Investment is allowed to depend on interest rate and output.
- We continue to assume that inventory investment is zero and that the economy is closed.

Investment, Sales, and the Interest Rate

5.1 The Goods Market and the IS Relation

- Let us relax the assumption that investment is exogenous.
- Investment depends on primarily on two factors:
 - ① The level of sales
 - ② The interest rate

$$I = I(Y, i) \quad (5.1)$$

(+, -)

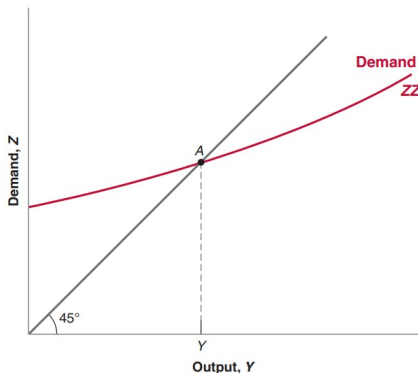
- With the revised investment function, the closed economy, goods market equilibrium condition becomes

$$Y = C(Y - T) + I(Y, i) + G \quad (5.2)$$

Determining Output

5.1 The Goods Market and the IS Relation

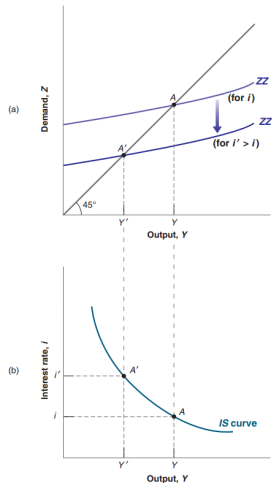
- Because we have not assumed that the consumption and investment are linear, ZZ is in general a curve rather than a line.



Deriving the IS Curve

5.1 The Goods Market and the IS Relation

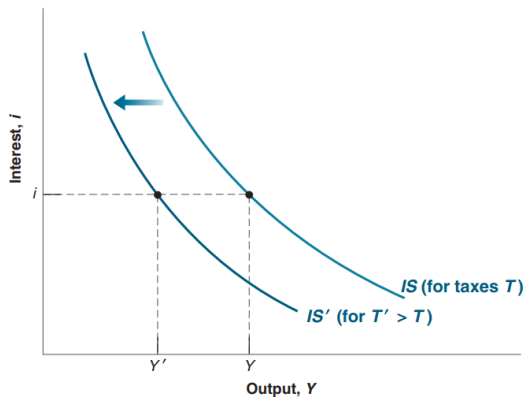
- To trace out an IS curve, start with a Keynesian cross with a **given interest rate**, then **change the interest rate**.



Shifts of the IS Curve

5.1 The Goods Market and the IS Relation

- Changes in either T or G will shift the IS curve.



Summary

5.1 The Goods Market and the IS Relation

- Downward-sloping IS curve: Equilibrium in the goods market implies that an increase in the interest rate leads to a decrease in output.
- Shifting the IS curve: Changes in factors that decrease (increase) the demand for goods given the interest rate shift the IS curve to the left (right).

Real Money, Real Income, and the Interest Rate

5.2 Financial Markets and the LM Relation

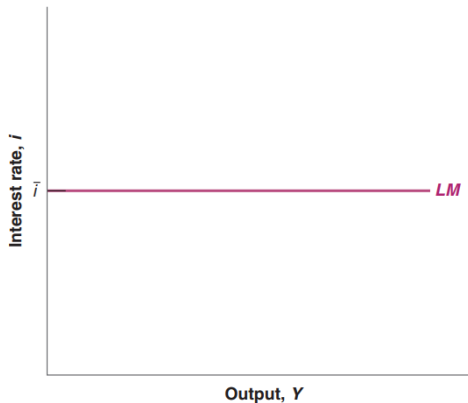
- Start with the money-market equilibrium condition from Chapter 4, rewrite nominal income as PY (where P is the price level), and divide by P to derive the real money market equilibrium condition:

$$\frac{M}{P} = Y \times L(i) \quad (5.3)$$

- Since the central bank chooses the interest rate (and adjusts the money supply so as to achieve it), the LM curve is graphed as a horizontal line.

Deriving the LM Curve

5.2 Financial Markets and the LM Relation



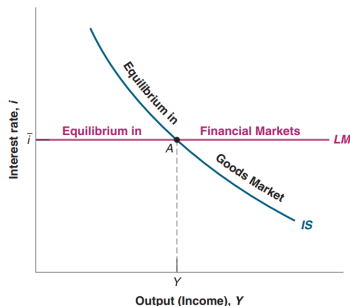
The $IS - LM$ Model

5.3 Putting the IS and LM Relations Together

- The IS relation follows from goods market equilibrium, and the LM relation follows from financial market equilibrium.
- They must both hold:

$$IS \text{ relation : } Y = C(Y - T) + I(Y, i) + G$$

$$LM \text{ relation : } i = \bar{i}$$



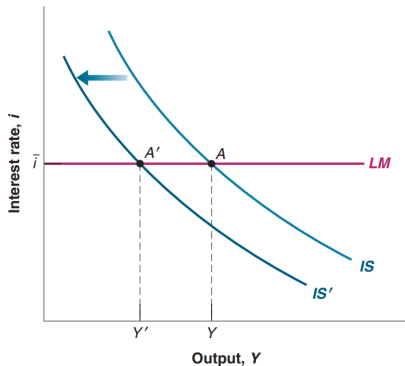
Comparative Statics Analysis

5.3 Putting the IS and LM Relations Together

- Steps for analyzing the effects of changes in policy or exogenous variables:
 - ① Does it shift the IS curve and/or the LM curve?
 - ② What does this do to equilibrium output and the equilibrium interest rate?
 - ③ Describe the effects in words.

5.3 Putting the IS and LM Relations Together

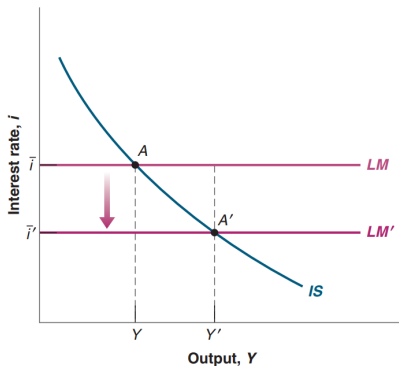
- Decrease in $G - T \Leftrightarrow$ **fiscal contraction** $\Leftrightarrow$ **fiscal consolidation**
- Increase in $G - T \Leftrightarrow$ **fiscal expansion**



Monetary Policy

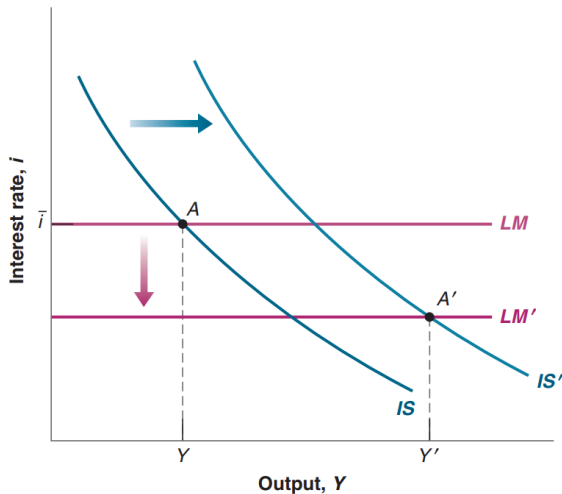
5.3 Putting the IS and LM Relations Together

- Decrease in $i \Leftrightarrow$ **monetary expansion**
- Increase in $i \Leftrightarrow$ **monetary contraction** $\Leftrightarrow$ **monetary tightening**



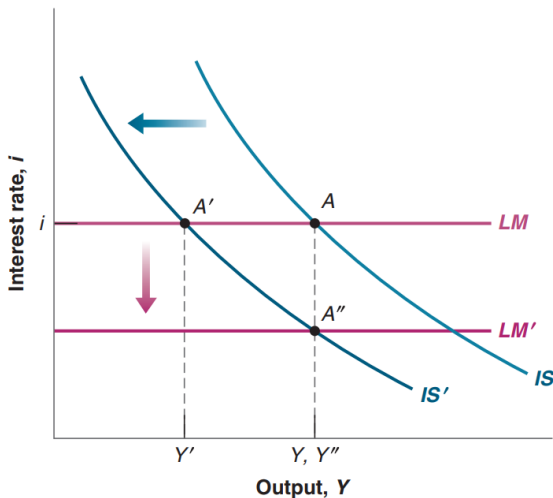
Combinations of Fiscal and Monetary Policy I

5.4 Using a Policy Mix



Combinations of Fiscal and Monetary Policy II

5.4 Using a Policy Mix



Speed of Adjustment

5.5 How Does the $IS - LM$ Model Fit the Facts?

- So far, we have ignored dynamics. As a matter of fact, it takes some time for consumption, investment, and output to adjust to an economic shock. How long is an empirical question.
- This question can be answered only by looking at the real data and using econometrics.
- Lawrence Christiano, Martin Eichenbaum, and Charles Evans, "The Effects of Monetary Policy Shocks: Evidence from the Flow of Funds," *Review of Economics and Statistics*, 1996, 78 (February): pp. 16-34.

The Empirical Effects of an Increase in the FFR

5.5 How Does the $IS - LM$ Model Fit the Facts?

